

PCB Defect Detection

Joint Multi-Domain Training — Runs 6 · 7 · 8

Abdullah Ashfaq | YOLOv11n | 200 epochs | June 2026

1. Experiment Overview

	Run6 (Baseline)	Run7 (Equal Data)	Run8 (BW Kaggle)
Training images	12 134	1 800	12 134
Kaggle images	8 534 (color)	900 (color, downsampled)	8 534 (binarized BW)
DeepPCB images	900 × 4 oversample	900 (no oversample)	900 × 4 oversample
Val set	Kaggle-val + DeepPCB 10%	Kaggle-val + DeepPCB 10%	Kaggle-val (BW) + DeepPCB 10%
Key variable	Strong Kaggle majority	Balanced domains	Domain appearance match

All three runs use YOLOv11n trained for 200 epochs (patience 50) with identical augmentation. The validation set never includes the DeepPCB test split, ensuring checkpoint selection is clean. The test mAP@50 numbers below are final held-out evaluations.

2. Overall Test Performance

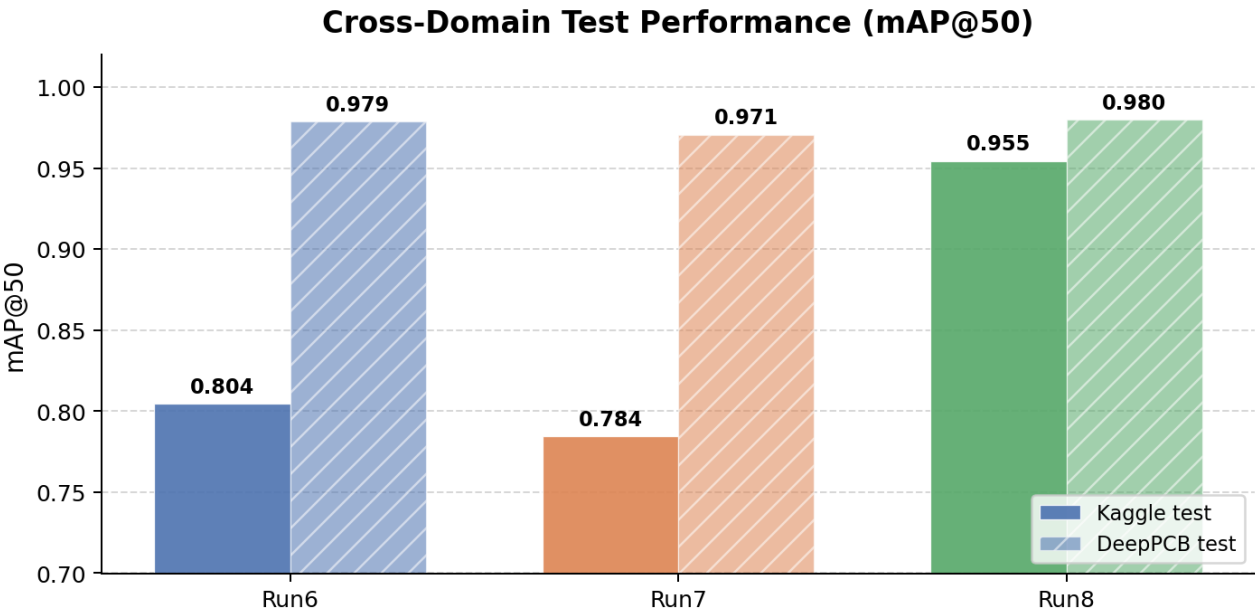


Figure 1. mAP@50 on held-out Kaggle test (solid) and DeepPCB test (hatched) sets. Run8's binarization strategy lifts Kaggle performance by +15 pp over Run6 while leaving DeepPCB accuracy virtually unchanged.

	Kaggle test mAP@50	DeepPCB test mAP@50	Δ Kaggle vs Run6	Δ DeepPCB vs Run6
Run6	0.8045	0.9788	—	—
Run7	0.7845	0.9707	−0.020	−0.008

Run8	0.9545	0.9798	+0.150 ▲	+0.001
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3. Per-Class Breakdown

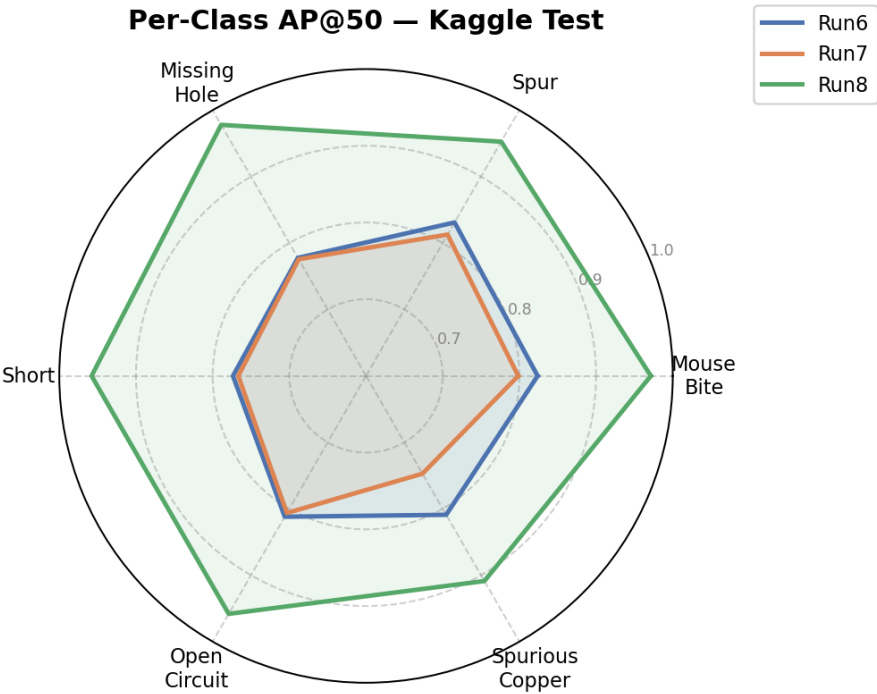


Figure 2. Radar chart of per-class AP@50 on the Kaggle test set. Run8 (green) consistently dominates all six defect categories. Run7 (orange) underperforms Run6 on spurious_copper, likely due to limited Kaggle training samples for that class.

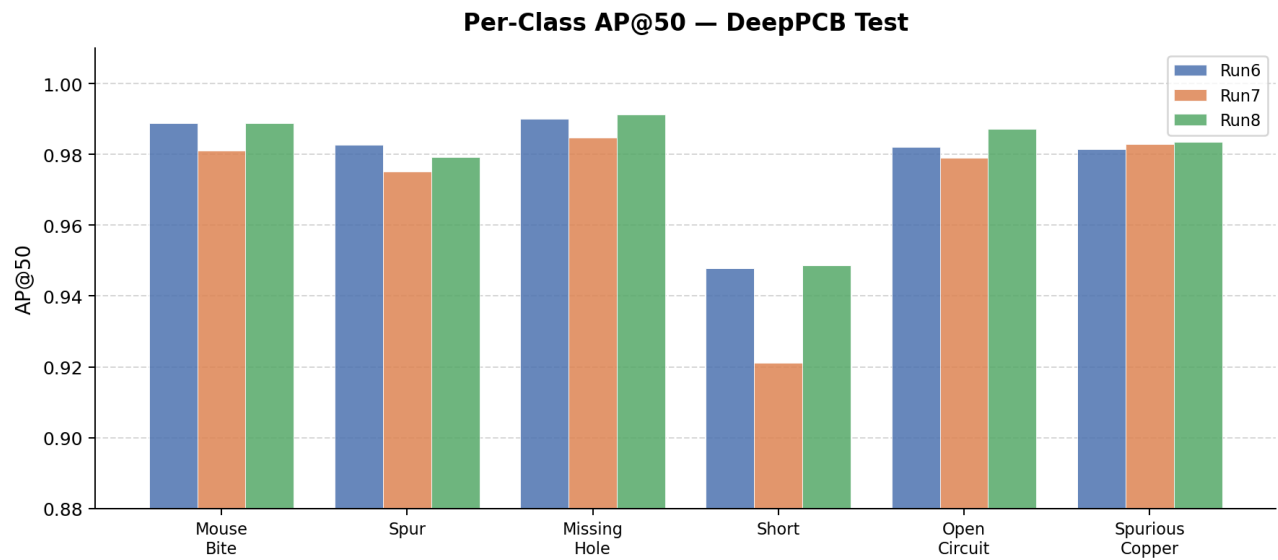


Figure 3. Per-class AP@50 on the DeepPCB test set. All three runs perform similarly (0.92–0.99), confirming that the DeepPCB domain is well-covered across strategies. The 'short' class remains the hardest for all models.

4. Training Dynamics

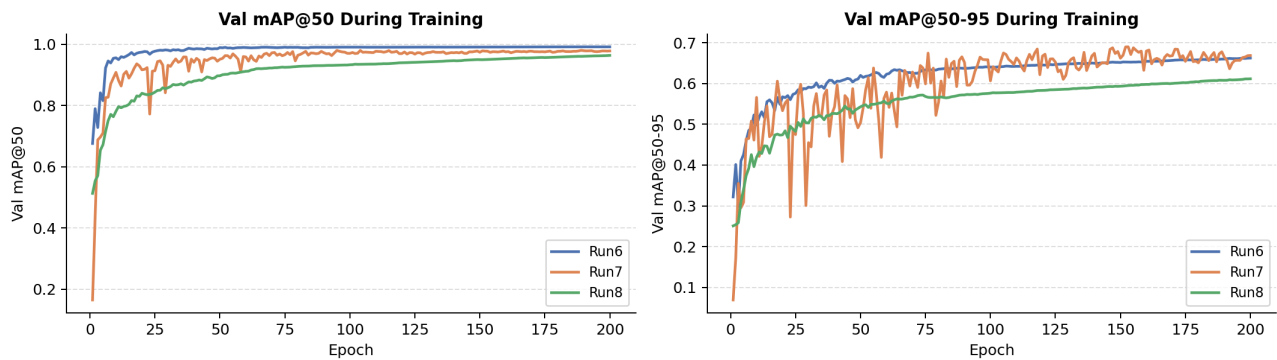


Figure 4. Validation mAP@50 and mAP@50-95 over 200 epochs (clean val set). Run6 and Run8 converge higher on mAP@50. Run7 converges faster but plateaus lower due to its smaller 1 800-image training set. Val metrics are on the mixed val set.

5. Per-Class Delta vs Run6

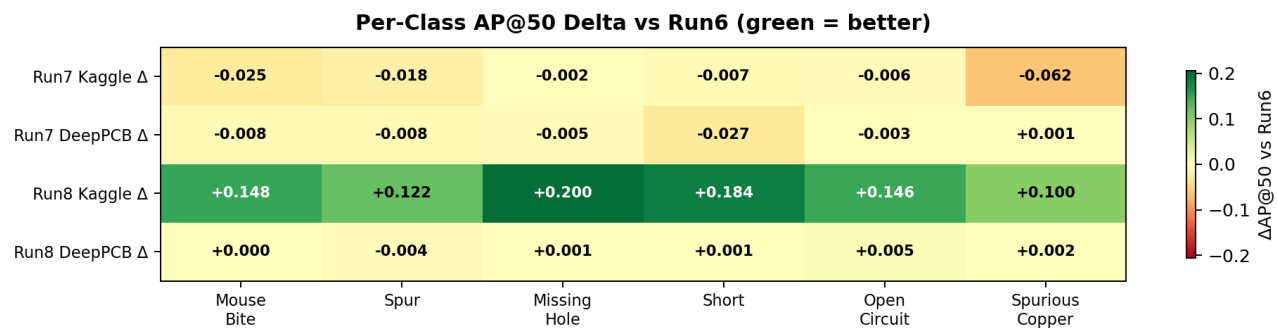


Figure 5. Per-class AP@50 change relative to the Run6 baseline. Green cells indicate improvement; red indicates regression. Run8 shows large green gains across all Kaggle classes (+8 to +19 pp). Run7 shows mild regressions on Kaggle, particularly spurious_copper (-6 pp).

6. Key Findings

- **Domain gap is the primary bottleneck.** Run8 shows that binarizing Kaggle images to match DeepPCB's grayscale appearance closes the cross-domain gap far more effectively than rebalancing data quantities.
- **Equal-data balancing (Run7) hurts.** Cutting Kaggle training data from 8 534 to 900 images starves the model on color-domain features, causing a -2 pp drop on Kaggle test with no benefit on DeepPCB.
- **DeepPCB is saturated.** All three runs score 0.97–0.98 on DeepPCB test, suggesting the model has learned the binary-image defect patterns well regardless of the Kaggle training strategy.
- **Run8 is the recommended checkpoint** for any deployment scenario that must handle both color PCB images and binary/grayscale inspection scans.